

MEDIA-BIAS ANALYSIS PLATFORM · ENGINEERING

Release Report

Reliability & Evaluation — what changed since the last production build

SCOPE	RANGE	DELTA	DATE
dev ☐ production	7ef7f3f ☐ oa28693	80 files · +6,671 / -575	16 June 2026

Executive summary

This release hardens the platform’s core promise — a trustworthy, defensible reliability score — and builds the measurement infrastructure to keep it honest. The headline is a correctness fix: confidence was being pinned to a misleading 65 / MEDIUM on well-sourced analyses because NLI “neutral” verdicts were wrongly counted as support failures. It now reads 95 / HIGH where the evidence warrants it. Alongside it: the post-analysis reliability layer was unified into one tested module, a full offline RAG-evaluation harness was added (with a path to real-data scoring), the classified source corpus grew from 34 to 55 outlets across the full political spectrum, and a complete brand asset kit was produced.

All work is on **dev** only; production remains on the rolled-back baseline and is untouched. The behavioural reliability change must be verified on staging before any production promotion.

12	623	7	55	0
COMMITTS	TESTS PASSING	NEW MODULES	CLASSIFIED OUTLETS	PROD CHANGES

☐ Headline fix — confidence accuracy (NLI “neutral”)

Symptom (observed on staging). A G7-summit analysis with 67 sources, all five camps covered (5/5) and every citation grounded (40/40) displayed Statement coverage 0%, dragging confidence to 65 / MEDIUM.

Root cause — two compounding bugs.

- “neutral” treated as failure. In NLI, neutral means the evidence neither supports nor contradicts a claim — i.e. unmeasured, not false. It was being counted as unsupported, so an analysis whose abstractive sentences all returned neutral scored 0%.
 - Single-article evidence. A synthesis sentence compresses facts from several articles, so no single best-match article entails it ☐ the judge returns neutral by default.
- Fix. “neutral” is now its own label, excluded from the support-ratio denominator (all-neutral ☐ unmeasured ☐ 0.85 factor, not 0). And each claim is judged against the union of its top-3 sources, so entailments actually land.

SCENARIO	STATEMENT COVERAGE	CONFIDENCE
Before (neutral counted as 0)	0%	65 · MEDIUM
After (all-neutral = unmeasured)	85% (unmeasured)	95 · HIGH
If top-K yields 70% real support	70%	90 · HIGH

IMPACT: CRITICAL	SCORE 9.5 / 10	+5 regression tests · directly protects the trust signal users see
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Changes by theme

□ Reliability orchestration unified

What. The post-analysis reliability layer (grounding □ dedupe □ blindspots □ NLI □ confidence □ coverage □ clusters) was extracted from inline server code into a single dependency-injected module, *composeReliability*. A duplicated, orphaned implementation was deleted.

Why it matters. Two sources of truth for the same logic inevitably drift — a direct risk to a “100% reliability” product. Pure steps are now imported; all I/O is injected, so the whole composition is unit-testable.

Impact. One source of truth; orchestration is now covered by 6 unit tests that previously could not run without a live server.

SCORE 9.0 / 10

+6 tests · dead module removed

□ Offline RAG-evaluation harness

What. A deterministic scorer (*lib/ragEval.js*) for retrieval (precision, recall@k, MRR, MAP, nDCG@k) and faithfulness (grounding × claim-support, with a contradiction penalty), run over a hand-labelled golden set of 15 German topics — each seeded to probe a specific failure mode.

Why it matters. Reliability was a claim, not a number. This makes it measurable and CI-gateable with no Gemini/DB in the loop, so regressions are caught before they ship.

Impact. Baseline established and gated (see scorecard). Surfaced the exact value/limits of lexical vs NLI verification.

SCORE 9.0 / 10

+ harness, golden set, metric tests

□ Real-corpus capture & scoring

What. A capture tool runs the exact production retrieval path against staging and dumps candidates per topic for human labelling; a *--real* mode then scores those labels with the same metrics. Captures include pre-gate rows, so labelling reveals both gate precision and gate recall.

Why it matters. The synthetic set tests logic, not reality. This is the bridge to measuring true accuracy on live German news, with independent (human) relevance judgement rather than self-authored fixtures.

Impact. Turns the eval from a logic check into a reality check; one-command capture via Railway, read-only and prod-safe.

SCORE 9.0 / 10

+ loader transform tests

□ Source corpus expansion — 34 □ 55 outlets

What. Added 21 classified outlets across all five camps (public broadcasters, major regionals, business/science, portals, and the fringe/extremist tail) on three axes — political spectrum, reach tier, factual rating — each with ownership and provenance.

Why it matters. No camp should be “thin”, or blind-spot detection fires on data gaps rather than real silence. Broader, honest coverage is the path to parity with comparable platforms.

Impact. Balanced breadth (8 / 11 / 11 / 12 / 13). Extremist outlets classed low-factual with citable basis; the coverage badge now derives the outlet count automatically.

SCORE 9.0 / 10

+5 integrity tests

□ Brand asset kit

What. Produced the six missing brand assets (icon, wordmark, social banners for X/LinkedIn/Instagram square & story) in the established editorial design, plus a single review PDF.

Why it matters. Consistent, ready-to-use brand presence across platforms; only the OG image existed before.

Impact. Complete kit in one place. Note: PDF pages are high-resolution raster renders, sufficient for review.

SCORE 8.0 / 10

visual review only

□ Carried quality fixes (this version)

What. Also included since the last production build: the relevance gate that removes off-topic filler, symmetric query embeddings (embed the topic, not a keyword bag), human-readable sub-story cluster labels (no LLM cost), and alignment of the reliability/insights panels with the editorial design system.

Why it matters. Each targets a visible accuracy or polish issue that affects perceived trust.

Impact. Off-topic articles no longer surface under unrelated topics; retrieval recall improved; UI is consistent.

SCORE 8.5 / 10

covered across suites

Quality scorecard

RAG evaluation — synthetic golden set, 15 topics (aggregate):

PRECISION	RECALL@5	MRR	MAP	nDCG@5	GROUNDING	CLAIM SUPPORT	FAITHFULNESS
97%	95%	100%	95%	97%	97%	96%	93%

All aggregate floors pass (precision ≥ 0.85 , recall@5 ≥ 0.80 , MRR ≥ 0.85 , faithfulness ≥ 0.60). Set-precision is the gated metric; p@5 is reported as a ranking diagnostic only.

Engineering scorecard:

AREA	SCORE	NOTE
Confidence / NLI accuracy fix	9.5 / 10	critical correctness, regression-tested
Reliability orchestration	9.0 / 10	single source of truth, fully tested
RAG eval harness (offline)	9.0 / 10	deterministic, CI-gateable

Real-corpus capture & scoring	9.0 / 10	bridge to real-data accuracy
Source expansion (34→55)	9.0 / 10	balanced, provenance-backed
Brand asset kit	8.0 / 10	raster PDF, review-grade
Overall release	9.0 / 10	reliability-focused, prod-safe

Rollout & risk

- Production untouched. All changes are on *dev*; production stays on the rolled-back baseline (0 prod changes).
- Verify before promotion. The confidence fix is behavioural — re-run the G7 query on staging and confirm HIGH, not 65.
- Tunable. The NLI evidence breadth (top-K) is a parameter; if staging still shows excess “neutral”, raise it from 3 to 4–5 with no rewrite.
- Next. Capture & label ~10 real topics to produce the first real-data reliability numbers and compare against the synthetic 97/95.

Prepared for the NeutralNews engineering team · generated from the dev branch diff against the production baseline.